

M T W T F S S

* CLEM-

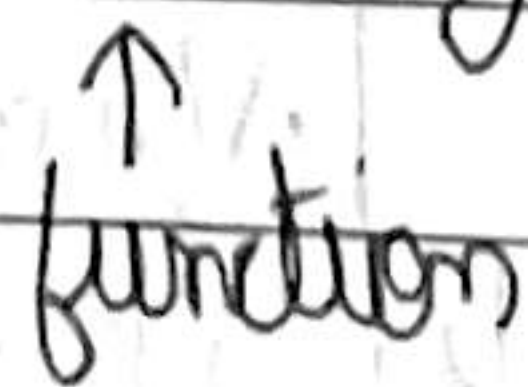
- ## # Reclassifying Fields -

- eg- If (depending-amount < 100 , "Low", If (depending-amount < 500 , "Medium", "High"))

- After deriving & reclassifying, it is crucial to validate the results to ensure accuracy & reliability.
- Involves verifying the new fields accurately.

- Modifies data values or creates new fields from one or more existing fields.

datetime-year (connect-date) = 2000 and handset = "ASAD"



↑
Field

↑
value

↑
Field

Variable	Value
T	

operator

operator

Expression Builder is a tool used to create or modify fields by writing expressions.

using functions, operators
& variables

Deriving Fields -

- Create new fields
- Use the derive node (field ops)
- Derive types

- Formula

- Flag

- Nominal

- Conditional

4) Math Functions -

- @ROUND(Salary/12,2)

- @LN(value)

Common CLEM functions -

1) Conditional expressions

↳ if(cond, true_val, false_val)

2) String functions

- @length(Name)

- @Substr(Name, 1, 3)

3) Date functions

- @DATEPART(Date, "year")

- @DATEDIFF(Date 1, Date 2, "days")

Reclassify node -

- Enables transformation from one set of categorical values to another
- Useful for ~~clustering~~ collapsing categories or regrouping data for analysis

Purpose & use case for CLEM

- 1) Data manipulation
- 2) Integration in software

Key Features of CLEM -

- 1) Expression handling
- 2) Scripting language
- 3) Control structures
- 4) Data type & handling
- 5) Integration with GUI

Syntax & Structure -

- 1) Commands & Functions
- 2) Variable
- 3) Conditional logic
- 4) Loops & Iteration

Applications

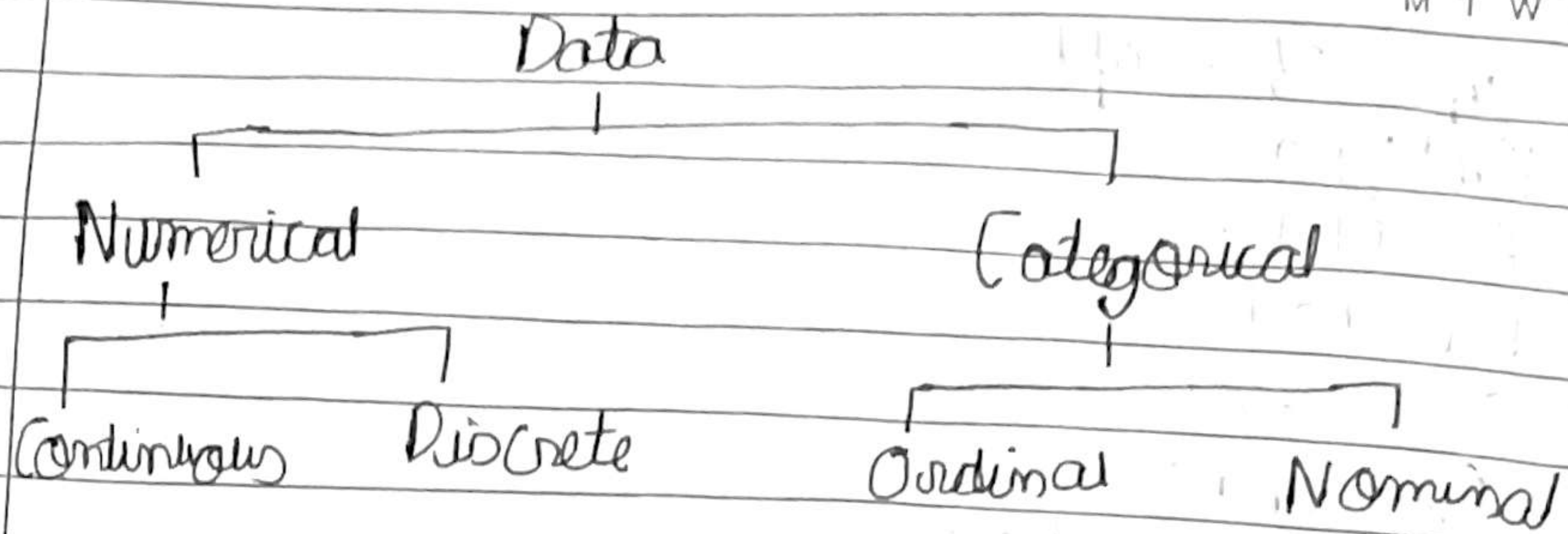
- 1) Data transformation
- 2) Rule based modeling
- 3) Automation

Advantages -

- 1) User friendly
- 2) Powerful & Flexible
- 3) Integration with data mining tools

Limitations -

- 1) Limited Scope
- 2) Performance
- 3) Learning curve



Data can be paired or unpaired -

- (i) Paired: Categories are related to one another
- (ii) Unpaired: Categories are not related to one another

Numerical data can be parametric or non parametric -

- (i) Parametric: approximately fits a normal distribution
- (ii) Non parametric: does not fit a normal distribution

For statistical test, we use 2 types of variables -

- (i) Independent variable
- (ii) Dependent variable

Statistical test parameters -

(i) Test statistic:

- Variable calculated from sample data & used in hypothesis test

- Used to determine whether a test was significant or not

(ii) Degrees of Freedom:

- No. of values of quantities that can be assigned to a statistical distribution

- Should be reported with test results

(iii) P-value/Level of significance:

- Measure of significance for the test statistic

- Typically 0.005

Types of Graphs -

- 1) Histogram
- 2) Box Plot
- 3) Bar chart
- 4) Scatter plot
- 5) Line charts
- 6) Multiple line charts
- 7) Pie ~~char~~ graph

Inferential Statistics -

Rule of analysis:

A. Explore your data

- Look at data
- Identify data
- Graph / describe data
- Formulate Question

B. Analyze your data:

- Set up hypothesis
- Check normality
- Select & run approx test

C. Interpret your results -

- Find test statistic, DF, and P-value
- Determine if significant
- State if null hypothesis rejected or not
- Write result
- Present approx plot

Tests for Normality -

1) Open ~~for~~ test for normality file in SPSS

2) For histogram:

└ Graphs → Chart Builder → Histogram

└ Variable under investigation to the horizontal

└ Select normal curve

3) QQ plot:

└ Analyze → Descriptive Statistics → Q-Q Plots

└ Can run both variables at the same time

4) Test for normality:

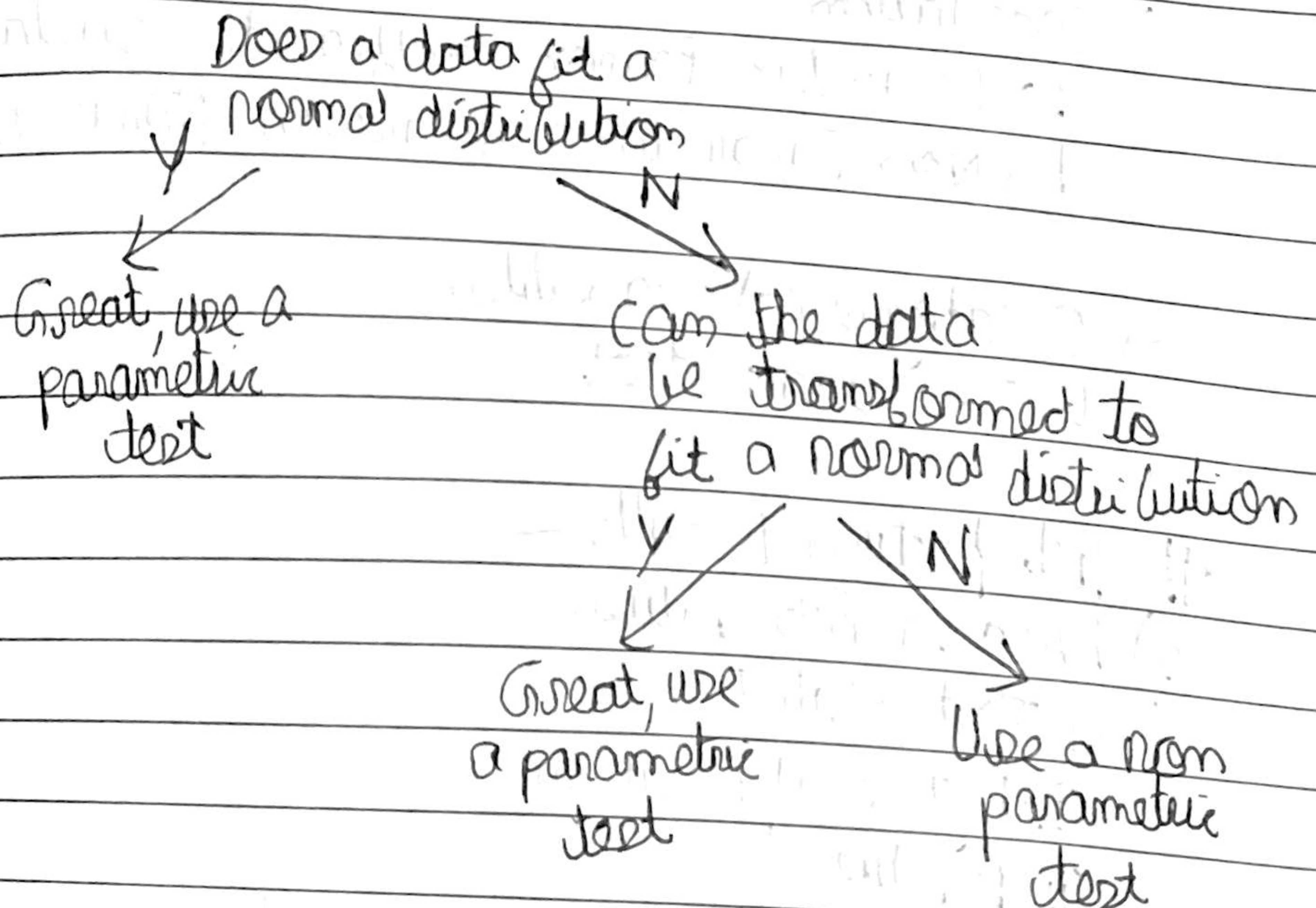
└ Analyze → Descriptive Statistics → Explore

└ Put variables to check under dependent list box

└ Click to select Normality plots with tests

* If data is not normally distributed, you can try transformations like Log transformation or Square root transformation

In SPSS: Transform → Compute variable



Paired = Dependent
2 sample = Independent

Quick Work

Page No.:

Date:

M T W T F S S

Types of tests -

- 1) 1 categorical variable + 1 numeric variable
 - Categorical variable is non paired & group number is one
 - One sample t-test
 - Categorical variable is non paired & group number is 2
 - Parametric: t-test
 - Non parametric: Mann Whitney Test
 - Categorical is paired & group number is 2.
 - Parametric: ~~ANOVA~~ Paired t-test
 - Non parametric: ~~Kruskal Wallis~~ Wilcoxon Signed Ranked Test
 - Categorical is non paired & group number > 2
 - Parametric: ANOVA
 - Non Parametric: Kruskal Wallace Test
 - Categorical is paired & group number > 2
 - Parametric: Repeated Measures ANOVA
 - Non Parametric: Friedman Test

2) 2 numerical variables Correlation

- Parametric: Pearson coefficient correlation
- Non Parametric: Spearman Rank Correlation

3) 2 categorical variables → Chi square test

Interpreting Results -

- 1) Find various values
 - (i) Test statistic
 - (ii) Degree of Freedom
 - (iii) P value
- 2) P value: tells whether to accept or reject H_0
 - (i) Typically, cutoff value is 0.05
 - (ii) If value < 0.05 , reject H_0
 - (iii) If value > 0.05 , fail to reject H_0

Reliability & Sensitivity

1) Reliability -

- Measures how consistent measurements are
- Inter-rater reliability deals with issue of reliability between different people (raters)
- Intra-rater reliability deals with whether one rater is consistent

2) Sensitivity -

- Measures how much measurements can detect true effects

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{FP + TN}$$

$$\text{Prevalence} = \frac{TP + FN}{TP + FN + FN + TN}$$

$$\text{Positive Predictive values} = \frac{TP}{TP + FP}$$

$$\text{Negative Predictive values} = \frac{TN}{FN + TN}$$

Odds Ratio -

- Measure of association between exposure & association outcome
- Odds that an outcome will occur given a particular exposure, compared to absence of that exposure

OR = 1 (exposure does not affect odds of outcome)

OR > 1 (exposure associated with higher odds of outcome)

OR < 1 (exposure associated with ~~higher~~ lower odds of outcome)

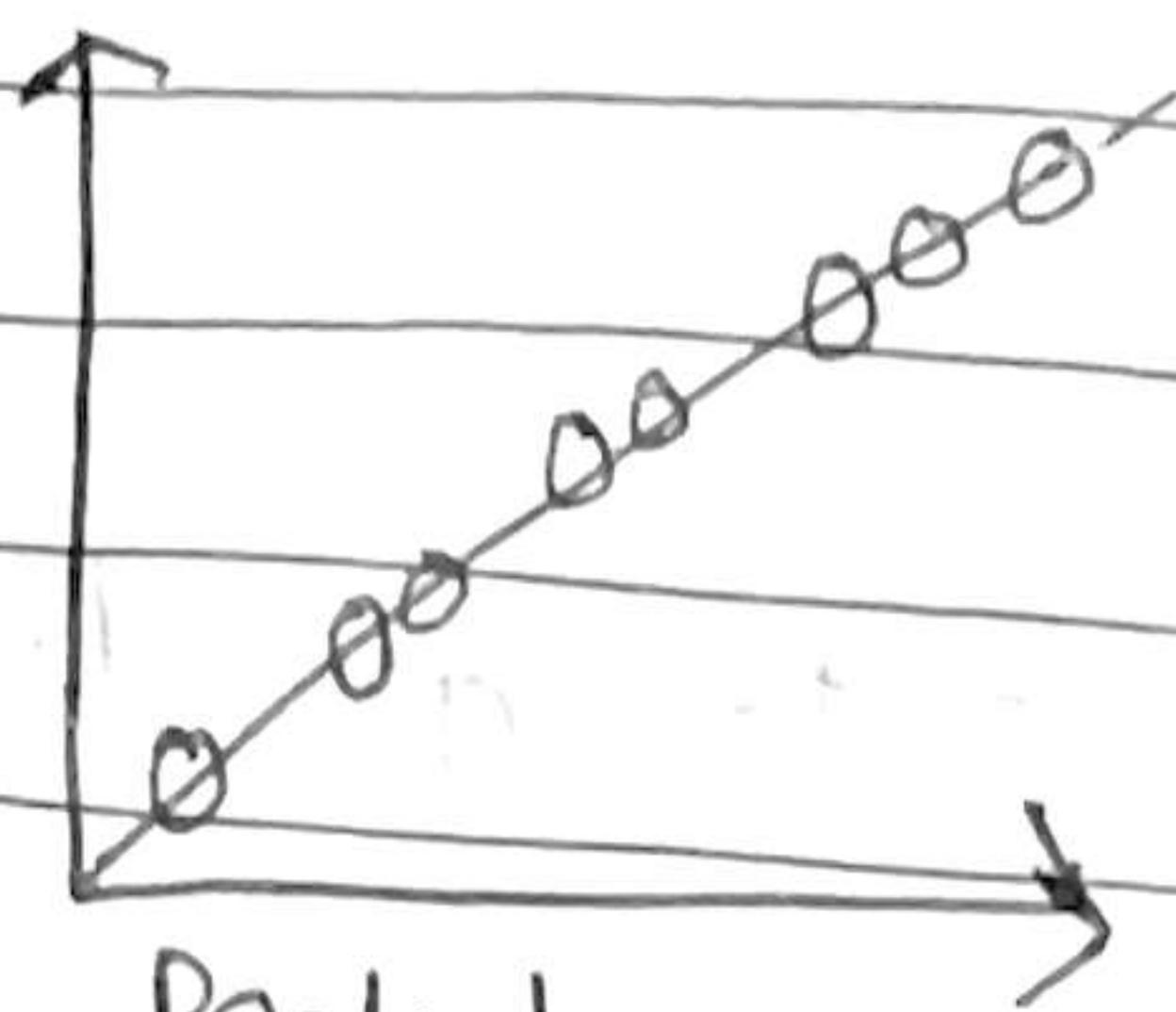
$$OR = \frac{ad}{bc}$$

a = No. of exposed cases
 b = No. of exposed non cases

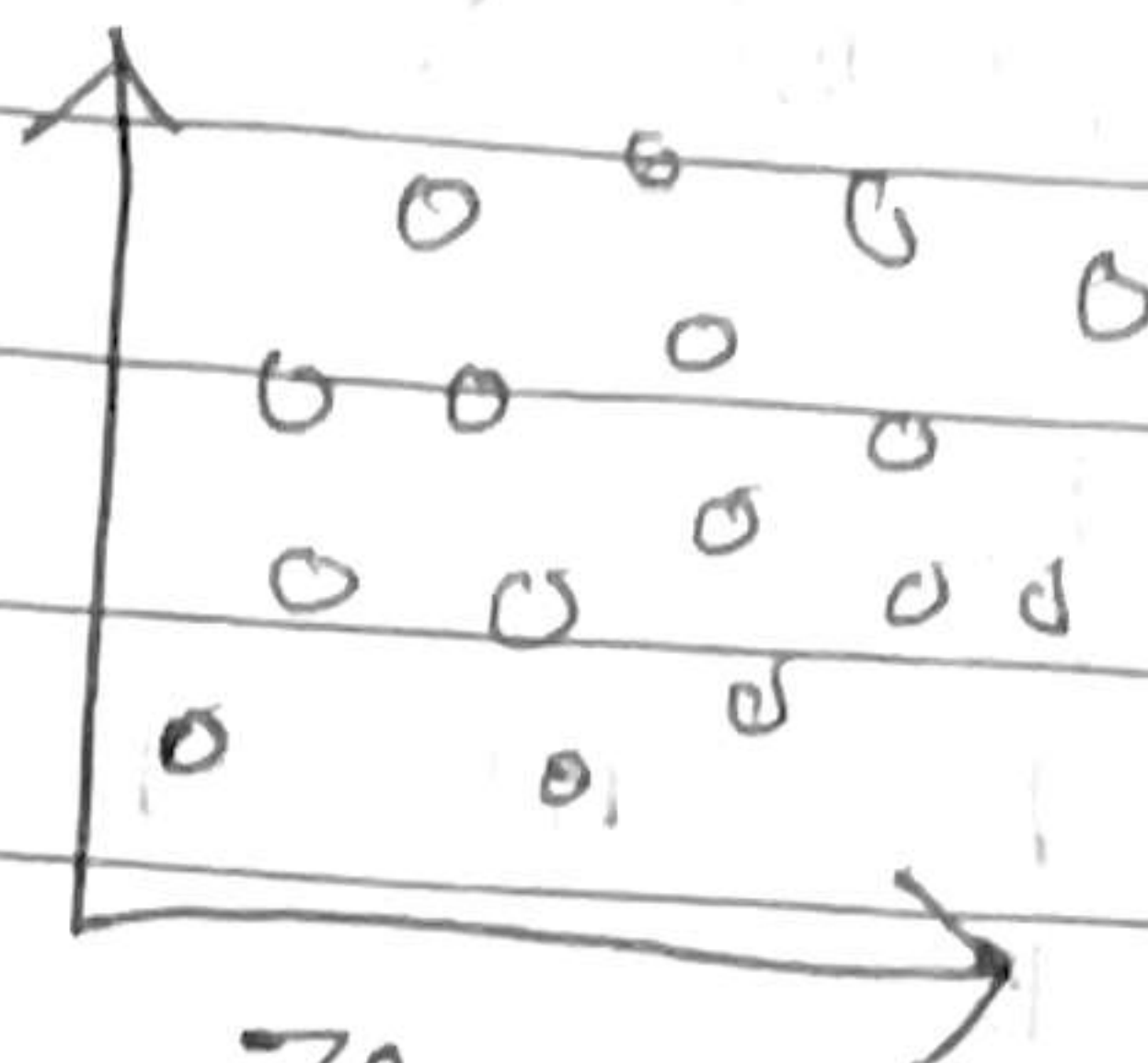
c = No. of unexposed cases

d = No. of unexposed non cases

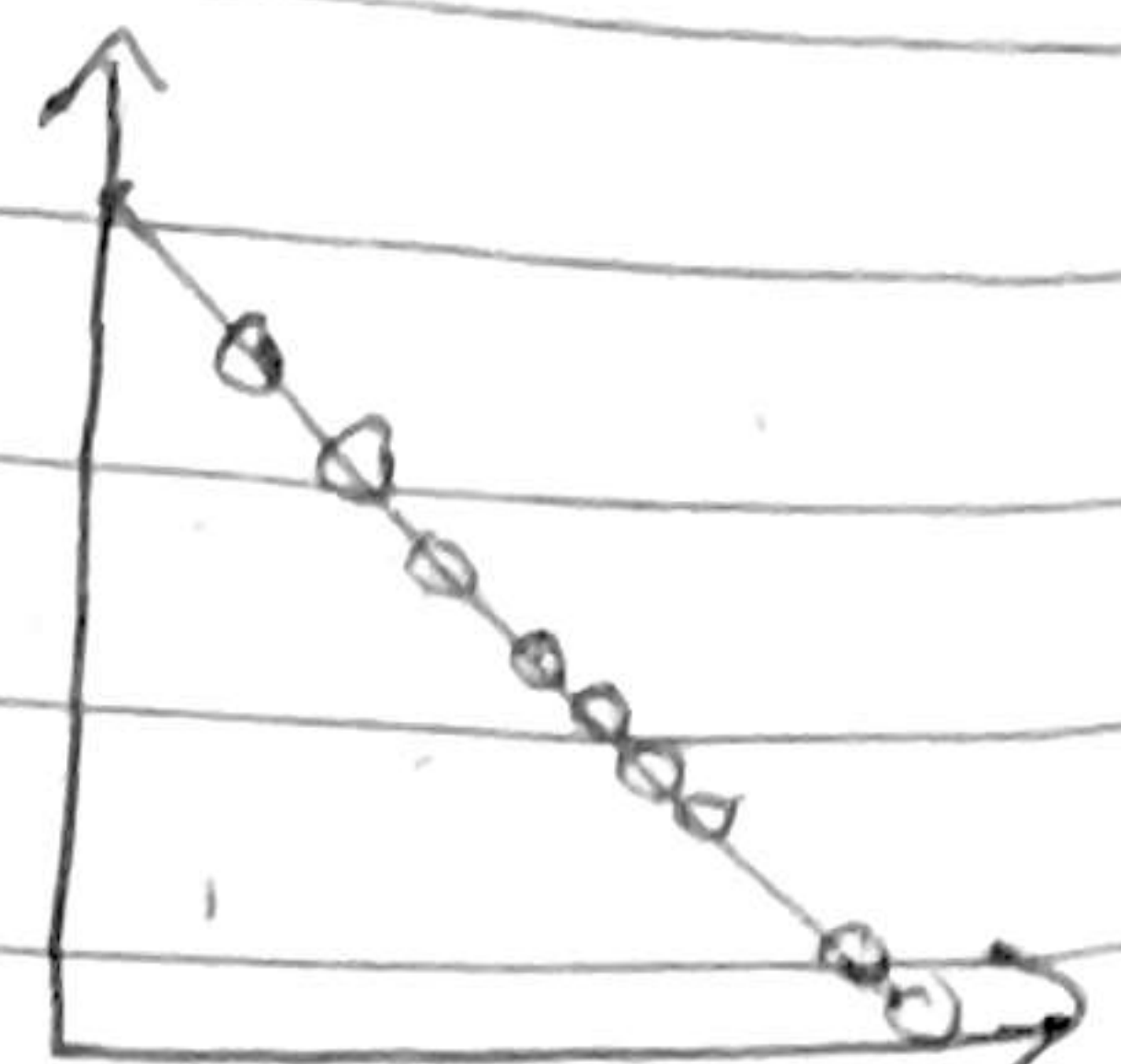
Correlation & Regression -



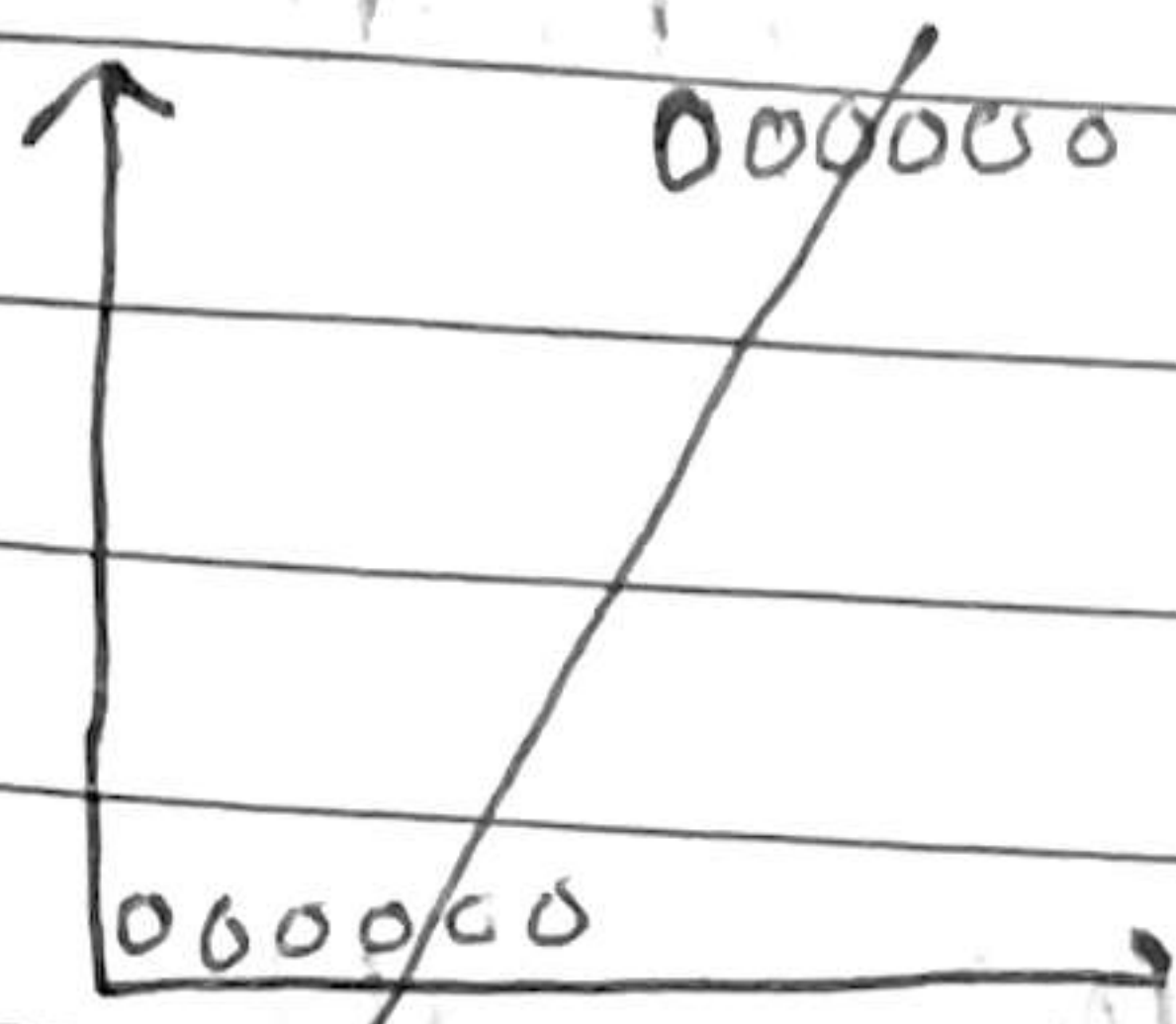
Perfect
Positive
Correlation



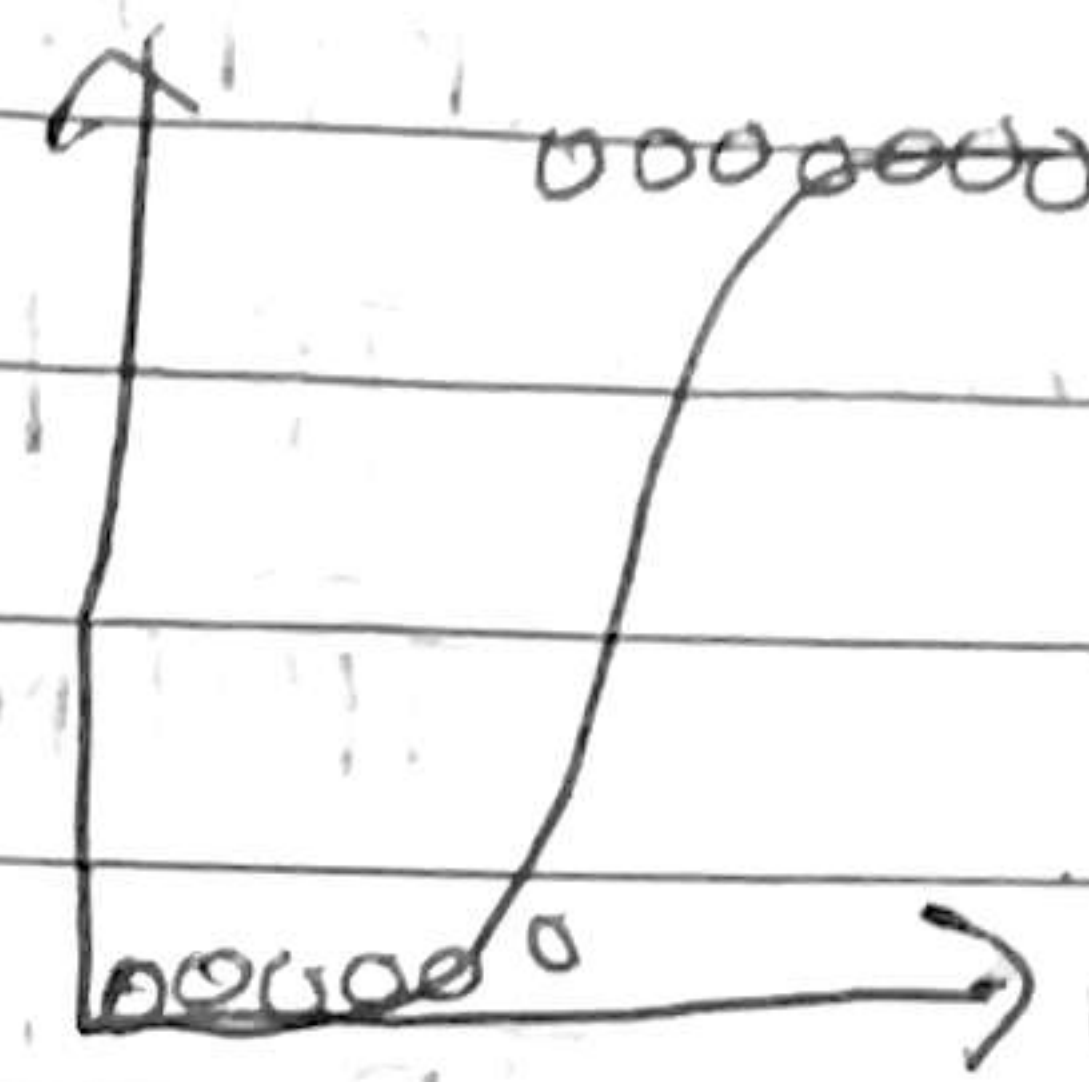
Zero
Correlation



Perfect
Negative
Correlation



Linear
Regression



Logistic
Regression

- Correlation = measures strength & direction of a linear relationship between 2 variables

- Regression = regression goes a step further than correlation. It helps you predict one variable based on the other

Best Fit Line -

Line with the lowest value of RSS would be the best fit line

$$RSS = \sum (Actual - Predicted)^2$$

	X	Y		
Q.	Actual	Predicted	(Residual)(X-Y)	(Residual) ²
	5	4.8	0.2	0.04
	7	7.2	-0.2	0.04
	6	5.9	0.1	0.01
	9	8.5	0.5	0.25
	10	9.2	0.3	0.09

$$RSS = 0.43$$

Distribution output = refers to the visualization of the frequency or spread of data points within a single field

Means output = provides average value of a field

Stats output = summarizes key metrics like mean, median, mode, variance, SD & percentile

$$\text{Mean } \bar{x} = \frac{\sum x_i}{n}$$

$$\text{Variance } S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \quad \text{SD } S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$\text{Mean absolute deviation } MAD = \frac{\sum |x_i - \bar{x}|}{n} \quad \text{Standard Error } SE = \frac{S}{\sqrt{n}}$$

Plot output = includes various types of graphs that visually represent the relationship between 2 or more fields

Data Transformations

1) Normalization

2) Binning Fields

3) Filling Fields

1) Normalization -

Rescales data to a range (usually 0 to 1)

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

2) Binning Fields -

Involves grouping continuous data into discrete bins or intervals

3) Filling Fields -

Involves handling missing data by substituting missing values with plausible estimates.

Techniques include imputation with mean, median & mode

Data Smoothing -

Used to reduce noise & variability in data, making trends, patterns, and signals more apparent.

Attribute Construction -

- Also known as Feature engineering

- Process where new attributes or features are created from existing data to improve performance of ML models.

Data Generalization -

Replacing specific values with more general categories or concepts.

Sequence data = where order matters

Quick Work

Page No:

Date:

M T W T F S S

Cross Record Functions -

- Involves operations that span multiple records or observations in a dataset.
- Allow you to compute aggregate values

Deriving a Counter Field -

- Involves creating a new variable or column in a dataset that counts occurrences or events
- Useful for various purposes such as tracking the frequency of certain activities

Restructuring data -

- Involves data transforming the format or layout of a dataset to make it more suitable for analysis or visualization
- May include tasks such as reshaping data from wide to long format or vice versa

Geospatial data -

- Involves info. related to geographic locations & time stamps

← Sample node

Sampling records -

- When we select a sample (smaller group / data) from a larger population => SAMPLING

- 2 primary types -

- (i) Probability sampling (involves random selection allowing strong statistical inference)
- (ii) Non Probability sampling (involves non random selection allowing you to easily collect data)

Population
= entire group

Sample
= specific group of individuals

Sampling frame -

- Actual list of individuals that the sample will be drawn from
- Ideally should include the entire target population
- Basically a list or database in the population from which a sample is actually drawn.

Types of Probability sampling methods -

1) Random sampling:

- ~~Gathering~~ Every ~~member~~ member of the population has an equal chance of being selected.
- Sampling frame should include the whole ~~attendant~~ population

2) Systematic sampling:

- Individuals from the population are chosen at regular intervals
- Eg: From a population (1 to 100)
Sample = (6, 16, 26, 36, ..., 96); $K = 10$
- Sampling interval = $\frac{\text{estimated population size}}{\text{sample size}}$
(k)

3) Stratified sampling:

- Involves dividing the population into subpopulations that may differ in imp ways.
- The divided sub groups are called strata.

4) Cluster sampling:

- Also involves dividing the population into sub groups, but each sub group should have similar characteristics to the whole sample
- You randomly select entire sub groups

- Good for dealing with large & dispersed populations but there is more risk of errors in the sample as there could be substantial differences between clusters.

Using Partitioning (Partition node) -

- Involves dividing a dataset into subsets based on specified criteria.
- Can help in managing & analyzing large datasets efficiently.

Balancing Records (Balance node)

- Refers to ensuring that the dataset used for analysis or modeling has a proportional representation of different classes or categories.
- Crucial in fraud detection.
- Techniques such as -
 - (i) Over sampling ($\uparrow$ the no. of minority class records)
 - (ii) Under sampling ($\downarrow$ the no. of majority class records)

Types of Non Probability sampling methods -

- Easier & cheaper to access
- Higher risk of sampling bias
- 1) Convenience sampling -
 - Includes individuals who happen to be most accessible to the researcher.
 - Can't provide generalizable results.
 - Risk of sampling bias & selection bias.

2) Voluntary response sampling -

- Based on ease of access
- Instead of researcher choosing participants, people volunteer themselves.
- Risk of self selection bias.

3) Purposive Sampling -

- Also known as judgement sampling
- Involves the researchers using their expertise to select a sample that is most useful to the purposes of the research.
- Often used in qualitative research.
- Risk of observer bias

4) ~~Sam~~ Snowball Sampling -

- Used to recruit participants via other participants
- Downside here is representativeness.
- Risk of sampling bias

5) Quota Sampling -

- Relies on non random selection of a predetermined number or proportion of units. This is called a quota.
- You first divide the population into mutually exclusive subgroups (called strata) & then reach your quota.

Using SQL (Structured Query Language) Pushback:

- Technique used in data integration & processing where some computation or filtering logic is delegated to the underlying database system through SQL queries.
- Reduces amount of data transferred between the database & application, improving overall performance.

Previewing SQL -

- Involves examining the results of a SQL query before executing it fully.
- Helps in optimizing queries

Identifying A-typical values-

- A-typical values = outliers
- Datapoints that ~~are~~ significantly ~~are~~ deviate from the majority of the data
- Techniques to detect outliers:
 - └ box plots
 - └ Z score analysis
 - └ domain knowledge

Processing missing values-

- Handling missing values:
 - └ removing records with missing values
 - └ imputing missing values using stats methods
 - └ ML algorithms

Global variable-

- Can be defined once & used across different parts of a program
- Help streamline coding by centralizing config or constants.
- Ensures consistency & ~~simplified~~ maintenance

Parameters-

- Variables that can be passed into functions, scripts, or queries to customize their behaviour or output.
- Enhance flexibility & reusability of code

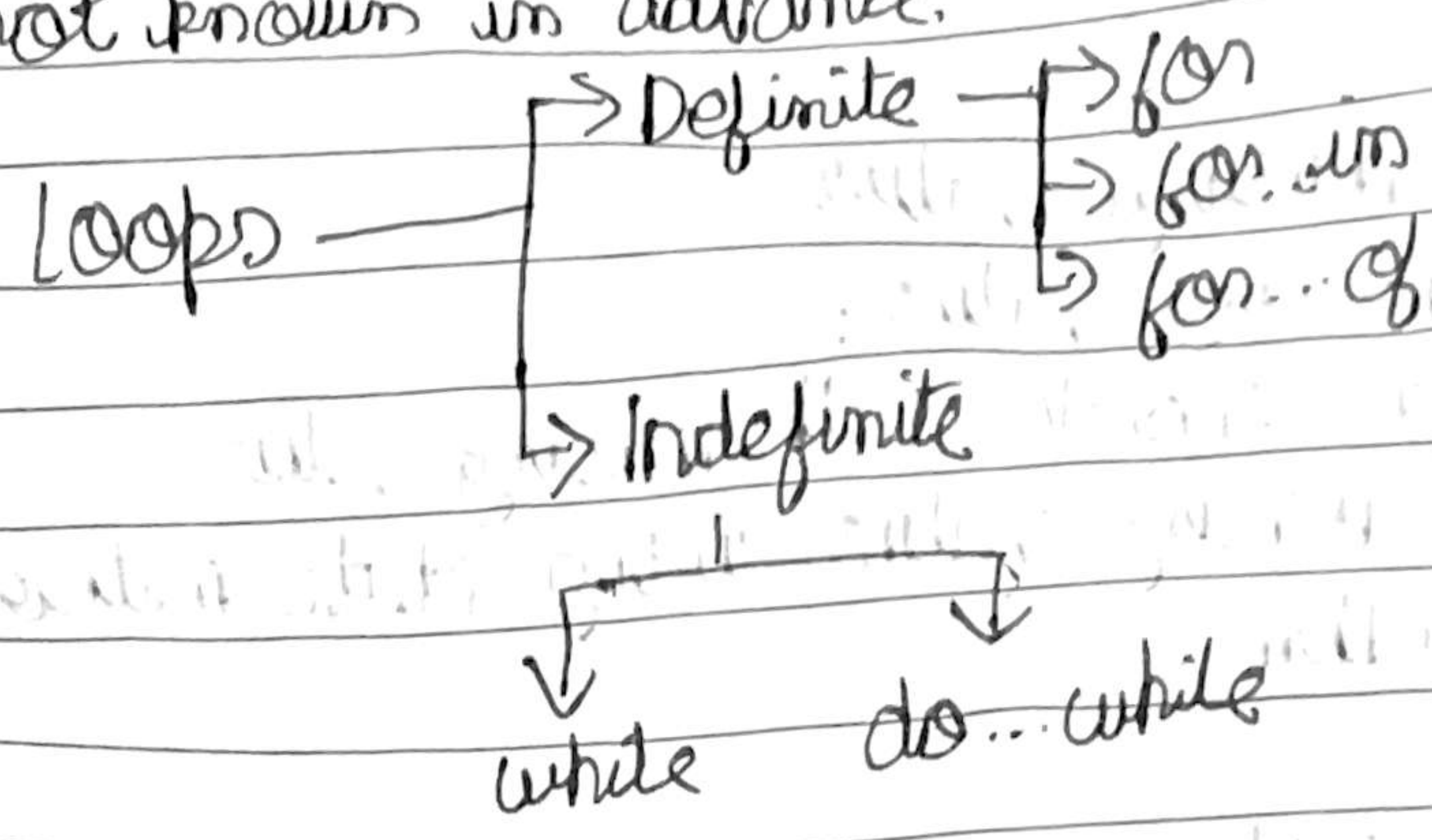
Conditional execution & looping-

- Programming constructs that allow ~~executions~~ executing specific blocks of codes based on certain conditions

Types of Loop

1) Definite looping -
 A block of code that is repeated a finite number of times, known in advance.

2) Indefinite looping -
 One where the no. of times its body repeats is not known in advance.



Feature	Purpose	Node/Path
1) SQL Pushback	Push operations to DB	Filter, Select nodes
2) Previewing SQL	Validate SQL before execution	Right click → click Preview SQL
3) A-Typical values	Detect outliers	Data Audit, Boxplot, Anomaly Detection
4) Processing Missing values	Identify nulls and gaps	Data Audit, Missing value node
5) Imputing Missing values	Fill missing data intelligently	Filler Node
6) Global	Session level constraints	Tools → Stream Properties → Global
7) Parameters	Reusable stream specific settings	Tools → Stream Properties → Parameters
8) Conditional Execution	Control flow execution	Condition node
9) Looping	Automate repetitive tasks	Loop node

Filler Node

- Used to handle missing values
- Replaces blanks or null values with
 - fixed constant
 - Statistical values (mean, median, mode)
 - Predicted values (from a model)

Aggregate Node

- Summarizes data & creates new fields at grouped levels

Type Node

- Not strictly for creating new fields, but crucial for field roles:
 - Input, Target, Partition, None
 - Measurement Levels (Continuous, Nominal, Ordinal)

Select Node

- Creates subsets based on conditions
- Indirectly defines new fields like "Selected = Yes/No"

Methods of Reclassification

- 1) Manual (define exact mappings for values)
- 2) Range based (continuous variables can be grouped into ranges)
- 3) Automatic (automatic based on data distribution by SPSS)
- 4) Lookup table (use an external file/table containing mapping rules)

Selecting a method to transform data -

- 1) Derive : Used to create new fields using expressions
Node on functions
- 2) Re-Classify : helps group or replace values for categorical
Node variables
- 3) Type : Modify metadata such as measurement level
Node
- 4) Select : Filters data records based on conditions
Node

How to use Binning Node in SPSS modeler -

- 1) Add a binning node (Field Ops)
- 2) Open the binning node
↳ select numeric field you want to bin
- 3) Choose binning methods
 - Automatic (equal width)
 - Equal frequency (quantile)
 - Manual
- 4) Apply & summarize